

Initial Parameter Configuration.

- CONTEXT_LENGTH: 30
- D_MODEL: 32
- *_LAYERS: 2
- *_HEADS: 4
- DROPOUT: 0.1
- LEARNING_RATE: 1e-4
- BATCH_SIZE: 64
- EPOCHS: 20 (with PATIENCE=10)

Experiment Name	`D_MODEL`	`*_LAYERS`	`*_HEADS`
Baseline (Small)	32	2	4
Medium	64	2	4
Large	64	4	8
X-Large	128	4	8

Small

--- Model Size & Regularization ---

D_MODEL = 32

ENCODER_LAYERS = 2

DECODER_LAYERS = 2

ENCODER_ATTENTION_HEADS = 4

DECODER_ATTENTION_HEADS = 4

DROPOUT = 0.1

Epoch 01/20 -> Train Loss: -1.26991204, Val Loss: -1.65727360

New best validation loss: -1.65727360. Saving model...

Epoch 02/20 -> Train Loss: -2.20737763, Val Loss: -2.21042946

New best validation loss: -2.21042946. Saving model...

Epoch 03/20 -> Train Loss: -2.50152380, Val Loss: -2.42937358

New best validation loss: -2.42937358. Saving model...

Epoch 04/20 -> Train Loss: -2.64044149, Val Loss: -2.35278372

Validation loss did not improve. Patience: 1/10

Epoch 05/20 -> Train Loss: -2.73635860, Val Loss: -2.59325069

New best validation loss: -2.59325069. Saving model...

Epoch 06/20 -> Train Loss: -2.79875934, Val Loss: -2.73347369

New best validation loss: -2.73347369. Saving model...

Epoch 07/20 -> Train Loss: -2.83933081, Val Loss: -2.61346611

Validation loss did not improve. Patience: 1/10

Epoch 08/20 -> Train Loss: -2.88090314, Val Loss: -2.80592543

New best validation loss: -2.80592543. Saving model...

Epoch 09/20 -> Train Loss: -2.90326997, Val Loss: -2.74745058

Validation loss did not improve. Patience: 1/10

Epoch 10/20 -> Train Loss: -2.93674289, Val Loss: -2.73024203

Validation loss did not improve. Patience: 2/10

Epoch 11/20 -> Train Loss: -2.95575961, Val Loss: -2.88569784

New best validation loss: -2.88569784. Saving model...

Epoch 12/20 -> Train Loss: -2.97721556, Val Loss: -2.72488405

Validation loss did not improve. Patience: 1/10

Epoch 13/20 -> Train Loss: -2.99184759, Val Loss: -2.90419203

New best validation loss: -2.90419203. Saving model...

Epoch 14/20 -> Train Loss: -3.01072159, Val Loss: -2.88882065

Validation loss did not improve. Patience: 1/10

Epoch 15/20 -> Train Loss: -3.02630497, Val Loss: -2.94505787

New best validation loss: -2.94505787. Saving model...

Epoch 16/20 -> Train Loss: -3.05087575, Val Loss: -2.88568141

Validation loss did not improve. Patience: 1/10

Epoch 17/20 -> Train Loss: -3.05573125, Val Loss: -2.92449716

Validation loss did not improve. Patience: 2/10

Epoch 18/20 -> Train Loss: -3.06843116, Val Loss: -2.97391046

New best validation loss: -2.97391046. Saving model...

Epoch 19/20 -> Train Loss: -3.07660289, Val Loss: -3.06566454

New best validation loss: -3.06566454. Saving model...
Epoch 20/20 -> Train Loss: -3.08992422, Val Loss: -2.98136848
Validation loss did not improve. Patience: 1/10

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Final Transformer Model Performance

Dataset Metric Value (EURUSD)

Test	RMSE	0.000320
Test	MAE	0.000207
Validation	RMSE	0.000405
Validation	MAE	0.000277
Training	RMSE	0.000400
Training	MAE	0.000268

Medium

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# --- Model Size & Regularization ---  
D_MODEL = 64  
ENCODER_LAYERS = 2  
DECODER_LAYERS = 2  
ENCODER_ATTENTION_HEADS = 4  
DECODER_ATTENTION_HEADS = 4  
DROPOUT = 0.1
```

Epoch: 100%

20/20 [3:15:50<00:00, 587.68s/it]

Epoch 01/20 -> Train Loss: -1.56692043, Val Loss: -2.04491891
New best validation loss: -2.04491891. Saving model...

Epoch 02/20 -> Train Loss: -2.48968552, Val Loss: -2.39814684
New best validation loss: -2.39814684. Saving model...

Epoch 03/20 -> Train Loss: -2.71723100, Val Loss: -2.56507115
New best validation loss: -2.56507115. Saving model...

Epoch 04/20 -> Train Loss: -2.83722939, Val Loss: -2.51692871
Validation loss did not improve. Patience: 1/10

Epoch 05/20 -> Train Loss: -2.90887464, Val Loss: -2.68082619
New best validation loss: -2.68082619. Saving model...

Epoch 06/20 -> Train Loss: -2.94121009, Val Loss: -2.62182762
Validation loss did not improve. Patience: 1/10

Epoch 07/20 -> Train Loss: -2.98727295, Val Loss: -2.69427769
New best validation loss: -2.69427769. Saving model...

Epoch 08/20 -> Train Loss: -3.00831732, Val Loss: -2.74463792
New best validation loss: -2.74463792. Saving model...

Epoch 09/20 -> Train Loss: -3.04316489, Val Loss: -2.72944727
Validation loss did not improve. Patience: 1/10

Epoch 10/20 -> Train Loss: -3.05835860, Val Loss: -3.00711411
New best validation loss: -3.00711411. Saving model...

Epoch 11/20 -> Train Loss: -3.08628419, Val Loss: -2.79829669

Validation loss did not improve. Patience: 1/10

Epoch 12/20 -> Train Loss: -3.10349345, Val Loss: -2.58897913

Validation loss did not improve. Patience: 2/10

Epoch 13/20 -> Train Loss: -3.11900397, Val Loss: -2.85472667

Validation loss did not improve. Patience: 3/10

Epoch 14/20 -> Train Loss: -3.12596807, Val Loss: -2.65825916

Validation loss did not improve. Patience: 4/10

Epoch 15/20 -> Train Loss: -3.15598823, Val Loss: -2.54860368

Validation loss did not improve. Patience: 5/10

Epoch 16/20 -> Train Loss: -3.15283559, Val Loss: -2.83547838

Validation loss did not improve. Patience: 6/10

Epoch 17/20 -> Train Loss: -3.15816795, Val Loss: -2.87471262

Validation loss did not improve. Patience: 7/10

Epoch 18/20 -> Train Loss: -3.17460733, Val Loss: -3.11084730

New best validation loss: -3.11084730. Saving model...

Epoch 19/20 -> Train Loss: -3.18600527, Val Loss: -2.83332502

Validation loss did not improve. Patience: 1/10

Epoch 20/20 -> Train Loss: -3.18009661, Val Loss: -2.95584041

Validation loss did not improve. Patience: 2/10

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      Final Transformer Model Performance
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Dataset Metric Value (EURUSD)
Test      RMSE      0.000383
Test      MAE       0.000282
Validation RMSE      0.000501
Validation MAE      0.000369
Training  RMSE      0.000453
Training  MAE       0.000317
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```

Large

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# --- Model Size & Regularization ---  
D_MODEL = 64  
ENCODER_LAYERS = 4  
DECODER_LAYERS = 4  
ENCODER_ATTENTION_HEADS = 8  
DECODER_ATTENTION_HEADS = 8  
DROPOUT = 0.1
```

Epoch 01/20 -> Train Loss: -1.56511191, Val Loss: -1.97137890
New best validation loss: -1.97137890. Saving model...

Epoch 02/20 -> Train Loss: -2.49864030, Val Loss: -2.46172267
New best validation loss: -2.46172267. Saving model...

Epoch 03/20 -> Train Loss: -2.72579328, Val Loss: -2.52139334
New best validation loss: -2.52139334. Saving model...

Epoch 04/20 -> Train Loss: -2.84819120, Val Loss: -2.77096074
New best validation loss: -2.77096074. Saving model...

Epoch 05/20 -> Train Loss: -2.92552035, Val Loss: -2.85414688
New best validation loss: -2.85414688. Saving model...

Epoch 06/20 -> Train Loss: -2.98155577, Val Loss: -2.87026985
New best validation loss: -2.87026985. Saving model...

Epoch 07/20 -> Train Loss: -3.02967374, Val Loss: -2.92231916
New best validation loss: -2.92231916. Saving model...

Epoch 08/20 -> Train Loss: -3.06231922, Val Loss: -2.92181175
Validation loss did not improve. Patience: 1/10

Epoch 09/20 -> Train Loss: -3.07958646, Val Loss: -3.02768169
New best validation loss: -3.02768169. Saving model...

Epoch 10/20 -> Train Loss: -3.11350687, Val Loss: -2.81074346
Validation loss did not improve. Patience: 1/10

Epoch 11/20 -> Train Loss: -3.12912381, Val Loss: -2.82500167
Validation loss did not improve. Patience: 2/10

Epoch 12/20 -> Train Loss: -3.14985260, Val Loss: -2.91157531
Validation loss did not improve. Patience: 3/10

Epoch 13/20 -> Train Loss: -3.16423515, Val Loss: -2.97330483
Validation loss did not improve. Patience: 4/10

Epoch 14/20 -> Train Loss: -3.18550154, Val Loss: -2.83880978
Validation loss did not improve. Patience: 5/10

Epoch 15/20 -> Train Loss: -3.18753895, Val Loss: -3.00616101
Validation loss did not improve. Patience: 6/10

Epoch 16/20 -> Train Loss: -3.20024961, Val Loss: -3.06356005
New best validation loss: -3.06356005. Saving model...

Epoch 17/20 -> Train Loss: -3.21648648, Val Loss: -3.09425579
New best validation loss: -3.09425579. Saving model...

Epoch 18/20 -> Train Loss: -3.22486945, Val Loss: -3.12669539
New best validation loss: -3.12669539. Saving model...

Epoch 19/20 -> Train Loss: -3.23186267, Val Loss: -3.22341946
New best validation loss: -3.22341946. Saving model...

Epoch 20/20 -> Train Loss: -3.24212141, Val Loss: -2.79386309
Validation loss did not improve. Patience: 1/10

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Final Transformer Model Performance			
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Dataset	Metric	Value	(EURUSD)
Test	RMSE	0.000323	
Test	MAE	0.000212	
Validation	RMSE	0.000393	
Validation	MAE	0.000259	
Training	RMSE	0.000377	
Training	MAE	0.000246	
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X-Large

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# --- Model Size & Regularization ---  
D_MODEL = 128  
ENCODER_LAYERS = 4  
DECODER_LAYERS = 4  
ENCODER_ATTENTION_HEADS = 8  
DECODER_ATTENTION_HEADS = 8  
DROPOUT = 0.1
```

Epoch 01/20 -> Train Loss: -1.90187958, Val Loss: -2.12124937
New best validation loss: -2.12124937. Saving model...

Epoch 02/20 -> Train Loss: -2.60081224, Val Loss: -2.55129800
New best validation loss: -2.55129800. Saving model...

Epoch 03/20 -> Train Loss: -2.75883833, Val Loss: -2.63368338
New best validation loss: -2.63368338. Saving model...

Epoch 04/20 -> Train Loss: -2.86316729, Val Loss: -2.49462560
Validation loss did not improve. Patience: 1/10

Epoch 05/20 -> Train Loss: -2.95045371, Val Loss: -2.85112942
New best validation loss: -2.85112942. Saving model...

Epoch 06/20 -> Train Loss: -2.98750535, Val Loss: -2.70208495
Validation loss did not improve. Patience: 1/10

Epoch 07/20 -> Train Loss: -3.05356066, Val Loss: -2.91829811
New best validation loss: -2.91829811. Saving model...

Epoch 08/20 -> Train Loss: -3.08695579, Val Loss: -2.95538808
New best validation loss: -2.95538808. Saving model...

Epoch 09/20 -> Train Loss: -3.11446021, Val Loss: -2.92007230
Validation loss did not improve. Patience: 1/10

Epoch 10/20 -> Train Loss: -3.14324193, Val Loss: -2.98325991
New best validation loss: -2.98325991. Saving model...

Epoch 11/20 -> Train Loss: -3.15723208, Val Loss: -2.95451909
Validation loss did not improve. Patience: 1/10

Epoch 12/20 -> Train Loss: -3.17723031, Val Loss: -2.91197978
Validation loss did not improve. Patience: 2/10

Epoch 13/20 -> Train Loss: -3.19683692, Val Loss: -3.03209658
New best validation loss: -3.03209658. Saving model...

Epoch 14/20 -> Train Loss: -3.21328322, Val Loss: -2.71035117
Validation loss did not improve. Patience: 1/10

Epoch 15/20 -> Train Loss: -3.22327348, Val Loss: -2.90121679
Validation loss did not improve. Patience: 2/10

Epoch 16/20 -> Train Loss: -3.23005403, Val Loss: -3.17755139
New best validation loss: -3.17755139. Saving model...

Epoch 17/20 -> Train Loss: -3.24474403, Val Loss: -3.04551603
Validation loss did not improve. Patience: 1/10

Epoch 18/20 -> Train Loss: -3.24573487, Val Loss: -3.20192347
New best validation loss: -3.20192347. Saving model...

Epoch 19/20 -> Train Loss: -3.25681419, Val Loss: -3.16360508
Validation loss did not improve. Patience: 1/10

Epoch 20/20 -> Train Loss: -3.26863132, Val Loss: -3.08302868
Validation loss did not improve. Patience: 2/10

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Final Transformer Model Performance			
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Dataset	Metric	Value	(EURUSD)
Test	RMSE	0.000326	
Test	MAE	0.000210	
Validation	RMSE	0.000415	
Validation	MAE	0.000284	
Training	RMSE	0.000427	
Training	MAE	0.000288	
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